PCA & Principal Component Analysis

Method Dossier with Field-Specific Examples | Standard Level

1. Method Description

Imagine: You have a dataset with 50 variables — sociodemographics, personality tests, health, consumer behavior. Many of these variables are correlated (e.g., age and health awareness). How do you find the underlying main dimensions? That is exactly what Principal Component Analysis (PCA) does.

PCA is a dimensionality reduction technique: it summarizes many observed variables into a few artificial dimensions (principal components) that capture as much of the original variance as possible. The first component explains the most variance, the second the second most (uncorrelated with the first), and so on.

Mathematically, PCA is based on the eigen decomposition of the correlation (or covariance) matrix. Eigenvalues (λ) indicate how much variance each component explains. Eigenvectors are the weights (loadings) of the original variables on the new components.

In short: PCA transforms many correlated variables into a few uncorrelated components — ideal for data visualization, noise reduction, and as a preprocessing step for further analyses (regression, cluster analysis).

Key Concepts

- Eigenvalue > 1 (Kaiser criterion): component explains more variance than a single variable
- Scree plot: the 'elbow' in the eigenvalue curve indicates the optimal number of components
- Cumulative variance: rule of thumb > 60–70 % for good representation
- Loadings: correlation between an original variable and a component (|loading| > 0.3 = relevant)
- Rotation (Varimax): simplifies interpretation by polarizing loadings
- Component scores: each observation receives a value per component

Example: Eigenvalue table from a PCA with 15 variables (n = 200). 4 components retained by Kaiser criterion.

Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1	4.82	32.1	32.1
PC2	2.94	19.6	51.7
PC3	1.73	11.5	63.2
PC4	1.21	8.1	71.3
PC5	0.88	5.9	77.2
...	...	...	...

2. Field-Specific Examples

Social Sciences

Question: Reduce 20 personality items (Big Five) to 5 dimensions. n = 1,000. PCA with Varimax rotation. Interpretation: Extraversion, Neuroticism, Conscientiousness, Openness, Agreeableness. Data source: SOEP / online panel.

Economics / Business

Question: 30 company performance indicators (revenue, profit, liquidity, debt, etc.) reduced to a few factors. PCA extracts 3 components: 'Profitability', 'Stability', 'Growth'. Cumulative variance: 74 %. Application: credit rating, benchmarking.

Health Sciences / Medicine / Psychology

Question: Reduce 40 neuropsychological test scores to core cognitive domains (memory, executive functions, processing speed). n = 500 (patients + controls). PCA serves as data reduction before cluster analysis to identify patient subgroups.

Humanities (Digital Humanities)

Question: Document-term matrix from a text corpus (500 texts, 2,000 word vectors). PCA reduces dimensionality for thematic mapping. The first two components visualize text similarities in 2D space. Application: stylometry, authorship analysis.

Sports Sciences

Question: 15 performance diagnostic tests (jump power, sprint time, endurance, flexibility, coordination) reduced to a few components. PCA extracts 3 components: 'Explosive strength', 'Base endurance', 'Coordination'. Enables efficient athlete classification (n = 200 athletes).

3. Assumptions and Prerequisites

- Metric variables (interval/ratio scale). Ordinal variables with 5+ levels can be used approximately
- Linear relationships between variables. PCA does not capture nonlinear structures
- Sufficient inter-variable correlation (KMO > 0.6; Bartlett test $p < 0.05$)
- No perfect multicollinearity (would yield singular matrix)
- Sample size: rule of thumb n > 5 × number of variables, better n > 10 × p
- Detect and handle outliers (PCA is sensitive to extreme values)

4. Interpretation and Reporting

A PCA is reported following this standard:

- Justification: why PCA? (dimensionality reduction, multicollinearity, exploration)
- Adequacy: report KMO (MSA) and Bartlett test
- Extraction: Kaiser criterion (EV > 1) + scree plot for component count decision
- Variance table: eigenvalues, variance explained (%), cumulative variance (%)
- Rotated loading matrix (e.g., Varimax): highlight loadings > 0.4, suppress < 0.3
- Name components: interpret meaning based on high-loading variables
- Optional: save component scores for further analysis

Report example:
'A PCA with Varimax rotation was conducted on 15 job satisfaction items ($n = 340$). KMO was 0.82; Bartlett test was significant ($\chi^2(105) = 1{,}892$, $p < 0.001$). Based on the Kaiser criterion, 3 components were extracted, jointly explaining 64.7% of total variance. Table 2 shows the rotated loading matrix. Component 1 ('Leadership', EV = 4.2) comprises supervisor behavior items, component 2 ('Colleagues', EV = 3.1) team climate, and component 3 ('Resources', EV = 2.4) work tools and workload.'

5. Code Examples

R — PCA with psych package:

```
library(psych)

# Extract PCA (5 components, Varimax)
pca_res <- principal(d, nfactors = 5, rotate = 'varimax', scores = TRUE)

# Display results
print(pca_res$loadings, cutoff = 0.3) # loadings
pca_res$values # eigenvalues
summary(pca_res$Vaccounted) # variance table

# Scree plot
scree(d, factors = FALSE)

# KMO & Bartlett
KMO(d)
cortest.bartlett(cor(d), n = nrow(d))

# Extract component scores
scores <- pca_res$scores
d$pc1 <- scores[, 1]
```

Python (scikit-learn) — PCA:

```
import pandas as pd
import numpy as np
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# Standardize (mandatory!)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(d)

# Run PCA
pca = PCA(n_components=5)
scores = pca.fit_transform(X_scaled)

# Results
print('Eigenvalues:', pca.explained_variance_)
print('Variance (%):', pca.explained_variance_ratio_ * 100)
print('Cumulative:', np.cumsum(pca.explained_variance_ratio_) * 100)

# Loadings
loadings = pca.components_.T * np.sqrt(pca.explained_variance_)
loadings_df = pd.DataFrame(loadings, columns=[f'PC{i+1}' for i in range(5)])
print(loadings_df)
```

SPSS — PCA (FACTOR):

```
* PCA with Varimax rotation
FACTOR
/VARIABLES item1 item2 item3 item4 item5
/MISSING LISTWISE
/ANALYSIS item1 item2 item3 item4 item5
/PRINT INITIAL CORRELATION DET KMO EXTRACTION ROTATION
/CRITERIA FACTORS(5) ITERATE(25)
/EXTRACTION PC
/CRITERIA ITERATE(25)
/ROTATION VARIMAX
/SAVE REG(ALL)
/METHOD=CORRELATION.

* Output: correlation matrix, KMO/Bartlett, eigenvalues,
* extracted variance, rotated component matrix,
* new variables FAC1_1 through FAC5_1 in data.
```

6. Extension: PCA vs. EFA — and When to Use Which

PCA and exploratory factor analysis (EFA) are often confused. The difference:

- PCA: Observed variables are combined into components that explain the total variance (including error variance). Goal: dimensionality reduction / data compression.
- EFA: Only the common variance is modeled; error variance is separated. Goal: uncovering latent constructs (theory-driven).

Rule of thumb: If you simply want to reduce dimensionality (e.g., for subsequent regression), use PCA. If you want to measure a theoretical construct (e.g., 'intelligence' or 'job satisfaction'), use EFA.

Varimax, Quartimax, Oblimin — Which Rotation?

- Varimax (orthogonal): components remain uncorrelated. Simplest interpretation.
- Quartimax: simplifies the rows of the loading matrix. Good if a general component is expected.
- Oblimin (oblique): components may correlate. More realistic but more complex.

PCA with Non-Metric Variables

For binary or ordinal data, specialized variants exist: categorical PCA (CATPCA in SPSS) or Multiple Correspondence Analysis (MCA). Standard PCA assumes metric, (approximately) normally distributed variables.

7. Exercises

Exercise 1: A dataset contains 12 items on study satisfaction (n = 300). KMO is 0.79. The first 5 eigenvalues are: 4.1 / 2.3 / 1.4 / 0.9 / 0.7. How many components would you extract by the Kaiser criterion? How much variance do they explain? (4.1 + 2.3 + 1.4 = 7.8 / 12 = 65 %)

Exercise 2: Run a PCA on the built-in iris dataset in R (4 metric variables, n = 150). How many components have EV > 1? Which variable loads most strongly on PC1? Visualize the first two components as a scatter plot (colored by Species).

Exercise 3: Why must data be standardized before PCA? What happens if you run PCA on the covariance matrix instead of the correlation matrix? Demonstrate with variables on different scales (height in cm, weight in kg, income in €).